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MLA0201 – Fundamentals of Machine Learning

Q1. Bayesian Networks for Predicting Machine Failures

Answer:

I would use a Bayesian Network (BN) to represent the relationships between machine conditions and machine failure.

- First, I would identify important variables such as temperature, vibration, pressure, oil level, and machine failure.
- These variables are represented as nodes in the Bayesian Network.
- Directed edges represent dependency between variables.
- For example, high temperature and high vibration may increase the probability of machine failure.
- I would collect historical machine data and calculate the required conditional probabilities.
- When new sensor data is received, Bayesian inference calculates the probability of failure.

For example:

Temperature → Machine Failure
Vibration → Machine Failure
Pressure → Machine Failure

If the system observes high temperature and abnormal vibration, the Bayesian Network can calculate:

$$P(F, ailure \mid TemperatureVibration)$$

If this probability is high, the system can generate an early warning.

Advantage: It can handle uncertainty and combine information from multiple sensors.

Conclusion: Therefore, Bayesian Networks are useful for predictive maintenance and early machine-failure detection.

Q2. MRF for Image Denoising

Answer:

I would use a Markov Random Field (MRF) because neighboring pixels in an image are usually related to each other.

- Each pixel can be represented as a node.
- Edges connect neighboring pixels.
- The observed noisy image is the input.
- The actual clean pixel values are treated as hidden or unknown variables.
- The MRF assigns higher probability to solutions where neighboring pixels have compatible values.
- It also considers how likely the clean image is given the noisy observation.

For example:

```
      Pixel
        |
Pixel — Pixel — Pixel
        |
      Pixel
```

The center pixel is influenced by its neighboring pixels.

The objective is to find the most probable clean image:

$$X^* = \arg\max_X P(X \mid Y)$$

where:

- X= clean image
- Y= noisy image

Advantage: MRF preserves important image structures while reducing random noise.

Conclusion: MRFs are effective for image denoising because they model spatial relationships between neighboring pixels.

Q3. HMM for Speech Recognition

Answer:

I would use a Hidden Markov Model (HMM) because speech is sequential data.

An HMM contains:

- Hidden states: phonemes, words, or speech units.
- Observations: audio features extracted from the speech signal.
- Transition probabilities: probability of moving from one speech state to another.
- Emission probabilities: probability of observing an audio feature from a particular state.

The process is:

Audio Signal

↓

Feature Extraction

↓

Observed Audio Features

↓

HMM

↓

Hidden Speech States

↓

Recognized Words

For example, when a person says "CAT", the audio signal is divided into feature sequences. The HMM estimates the most probable sequence of hidden phonemes.

The Viterbi algorithm can be used to find the most likely hidden-state sequence.

Advantage: HMMs are suitable for sequential signals where the current state depends on previous states.

Conclusion: Therefore, HMMs can convert a sequence of audio observations into the most probable sequence of speech units or words.

Q4. CRF for POS Tagging

Answer:

A Conditional Random Field (CRF) is a probabilistic model used for sequence labeling.

In Part-of-Speech (POS) tagging, the input is a sentence and the output is a POS tag for every word.

Example:

Input:

The dog runs

Output:

The → Determiner

dog → Noun

runs → Verb

CRF considers the entire input sequence when predicting the output labels.

It can consider features such as:

- Current word
- Previous word
- Next word
- Word suffix
- Capitalization
- Previous POS tag

The CRF estimates:

$$P(Y \mid X)$$

where:

- X= input words
- Y= sequence of POS tags

Advantage: It considers relationships between neighboring labels and can use many overlapping features.

Conclusion: CRFs are effective for POS tagging because they predict the entire label sequence while considering useful features from the input sentence.

Q5. Conditional Independence in Probabilistic Inference

Answer:

Conditional independence means that two variables become independent when we know the value of another variable.

A simple graphical model is:

Disease

/ \

Fever Cough

Here, Fever and Cough may be dependent because both are caused by Disease.

But if we already know whether the patient has the disease, the relationship between Fever and Cough can become conditionally independent:

$$P(F, everCough \mid Disease) = P(Fever \mid Disease)P(Cough \mid Disease)$$

Without conditional independence, we may need to calculate a large joint probability distribution.

With conditional independence, the problem can be divided into smaller calculations.

Advantage:

- Reduces computational complexity.
- Requires fewer probabilities.
- Makes inference faster.
- Makes graphical models easier to understand.

Conclusion: Conditional independence simplifies probabilistic inference by allowing a large probability problem to be broken into smaller independent components.

Q6. Directed vs Undirected Graphical Models

Answer:

The two important types are Directed Graphical Models and Undirected Graphical Models.

Directed Graphical Model — Bayesian Network

Uses directed arrows.

Disease → Fever

Disease → Cough

It represents conditional dependencies using direction.

Undirected Graphical Model — Markov Random Field

Uses connections without direction.

Fever — Disease — Cough

It represents relationships between variables without specifying a causal direction.

Feature	Bayesian Network	MRF
Graph	Directed	Undirected
Direction	Yes	No
Common use	Diagnosis, prediction	Image processing
Probability	Conditional probabilities	Potential functions
Example	Disease → Symptom	Neighboring pixels

For healthcare diagnosis, I would prefer a Bayesian Network when causal relationships are important.

Conclusion: Bayesian Networks are useful for directional/causal relationships, while MRFs are useful when relationships are naturally symmetric.

Q7. Bayesian Inference with Missing Patient Data

Answer:

If some patient symptoms are missing, Bayesian inference can still make a diagnosis by using the available evidence.

Suppose we have:

Disease

/ | \

Fever Cough Fatigue

If the patient's cough information is missing but fever and fatigue are available, we can calculate:

$$P(D, isease \mid FeverFatigue)$$

The missing symptom can be treated as an unobserved variable and marginalized out.

In general:

$$P(D \mid E) = \frac{P(D, E)}{P(E)}$$

where Erepresents the available evidence.

The system:

1. Takes available symptoms.
2. Ignores or marginalizes missing symptoms.
3. Calculates posterior probabilities.
4. Compares the probabilities of possible diseases.
5. Selects the most probable diagnosis.

Advantage: Bayesian Networks can naturally handle uncertain and incomplete information.

Conclusion: Missing patient data does not necessarily prevent diagnosis because Bayesian inference can calculate probabilities using the available evidence.

Q8. HMM for Customer Purchase Sequences

Answer:

I would use an HMM to model customer purchasing behavior over time.

The customer's actual preference or shopping intention can be considered a hidden state, while purchases are the observations.

For example:

Hidden States:

Browsing → Interested → Ready to Buy

Observations:

View Phone → View Laptop → Add Laptop → Purchase

The HMM contains:

- States: Customer interests.

- Observations: Product views, clicks, cart additions, and purchases.
- Transition probabilities: Probability of changing from one interest state to another.
- Emission probabilities: Probability of a customer action given a particular state.

The system can estimate the customer's current hidden state and predict the next likely action.

For example, if a customer repeatedly views laptops and adds one to the cart, the system may predict a high probability of purchase.

The recommendation system can then recommend:

- Laptop accessories
- Similar laptops
- Laptop offers

Conclusion: HMMs help recommendation systems understand sequential customer behavior and predict future actions.

Q9. Why CRF Often Outperforms HMM in Named Entity Recognition

Answer:

Both HMM and CRF can be used for sequence labeling, but CRFs often perform better for Named Entity Recognition (NER).

Consider:

Anand works at Google in Chennai.

NER should identify:

Anand → PERSON

Google → ORGANIZATION

Chennai → LOCATION

HMM

HMM mainly models:

$$P(X, Y)$$

It makes strong independence assumptions about observations and labels.

CRF

CRF directly models:

$$P(Y \mid X)$$

It can use many features of the complete input sentence.

For example:

- Capitalization
- Prefix and suffix
- Previous word
- Next word
- Word position
- Neighboring labels
- Context words

Therefore, CRFs can use rich overlapping features without requiring them to be independent.

Conclusion: CRFs often outperform HMMs in NER because they consider the entire input context and multiple useful features while predicting the label sequence.

Q10. Graphical Model Learning and Generalization for Fraud Detection

Answer:

Graphical models can represent relationships between different variables involved in fraudulent transactions.

For example:

User

↓

Transaction → Fraud

↑ ↑

Location Device

Important variables can include:

- Transaction amount
- Location
- Device
- Time

- User behavior
- Transaction frequency

The model learns the relationships from historical transaction data.

For example, if a transaction occurs from an unusual location using a new device with an unusually large amount, the model can estimate a higher probability of fraud.

Learning

The model learns probabilities or parameters from training data.

Generalization

Generalization means the model should perform well on new transactions, not only on the training data.

To improve generalization:

- Use representative training data.
- Avoid overfitting.
- Use appropriate model complexity.
- Validate using unseen data.
- Include important fraud patterns.

Graphical models also allow relationships between multiple variables to be represented instead of looking at each feature independently.

Conclusion: Graphical model learning identifies important relationships in transaction data, while good generalization ensures that the learned fraud patterns work accurately on new and unseen transactions.